Dissimilarity Measures for Probability Distributions with an Application in Music Artist Classification

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Data

We used data sampled from PQRC Mentors and Math, Statistics and Computer Science Department Faculty.

- Individuals were asked to list 3 to 5 of their favorite bands/artists.
- This gave a data set of 95 musical artists.
- Musical characteristic statistics from every song of each artist were taken from the spotifyr package.

Motivation

Our goal was to perform hierarchical clustering on musical artists based on the characteristics of each of their songs. However, traditional approaches would overweight artists with more songs released. In order to give equal weight to all artists, we modelled each artist's style as a multivariate normal distribution. We then calculated dissimilarities between these distributions for hierarchical clustering.

Musical Artist Style Representation

The procedure for representing a musical artist's style as a multivariable normal distribution is as follows:

- Standardize song characteristics, so that all predictor variables are on the same scale.

$$x'_{i,j} = \frac{x_{i,j} - \overline{x_j}}{\sigma_j}$$

- Calculate mean vectors for each artist's standardized song characteristics, μ_k
- Calculate variance-covariance matrices for each artist's standardized predictor variables, Σ_k

Now we can model the k-th artist's music style as the multivariate normal distribution:

$$f_k \sim N(\mu_k, \Sigma_k)$$

Dissimilarity Measures

Let f_j and f_k be probability density functions. Then we define the following dissimilarity measures:

Squared Hellinger Distance:

$$H^2(f_j, f_k) = \frac{1}{2}\int\left(\sqrt{f_j} - \sqrt{f_k}\right)^2$$

KL Divergence:

$$D_{KL}(f_j \mid f_k) = \int_{-\infty}^{\infty} \ln\left(\frac{f_j(x)}{f_k(x)}\right) f_j(x)\,dx$$

Note, KL Divergence is not symmetric. To resolve this, we use the average KL Divergence:

$$D_{KL}(f_j, f_k) = \frac{1}{2}D_{KL}(f_j \mid f_k) + \frac{1}{2}D_{KL}(f_k \mid f_j)$$

Bhattacharyya Distance:

$$D_B(f_j, f_k) = -ln\left(\int_{-\infty}^{\infty}\sqrt{f_j(x)f_k(x)}\,dx\right)$$

Hierarchical Clustering

We calculated the dissimilarity of each artists using the three previously mentioned measures. Clustering was performed using the hclust package in R.

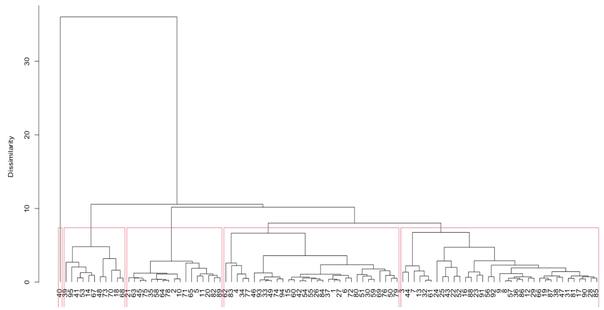

Figure 1: Dendrogram for Hierarchical Clustering using the Bhattacharyya Distance with complete linkage. Note that artist 40 (Joni Mitchell) is extremely dissimilar from all other artists.

Observations

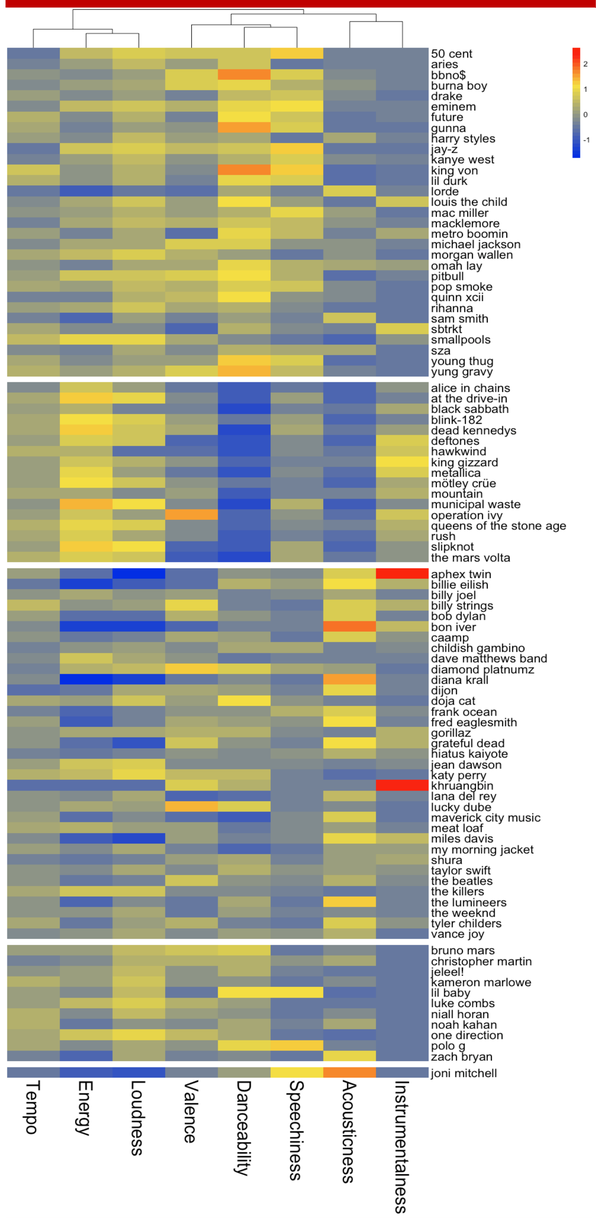

Figure 2: Average Standardized Song Characteristics of Music Artists. Artists are Grouped by Cluster Membership from Bhattacharyya Distance Hierarchical Clustering with 5 Clusters.

Analyzing Clusters

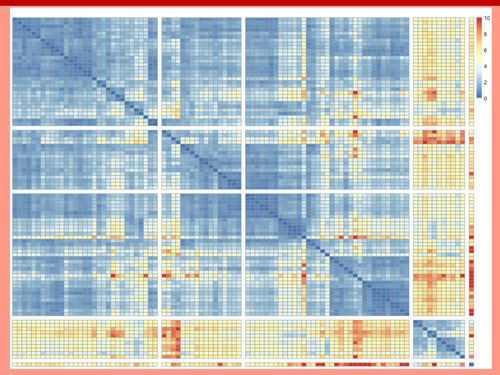

Figure 3: Heat Map of Bhattacharyya Distances Between Every Musical Artist.

- Observations are sectioned into 5 clusters, based on hierarchical clustering
- Blocks on the diagonal show distances between observations in the same cluster
- Off-diagonal blocks show distances between observations in different clusters.

Analysis and Conclusions

- The five clusters generated by Bhattacharyya distance with Hierarchical clustering are not dense
  - ❖ Other linkages for hierarchical clustering may produce more desirable results (we used "*complete*" linkage)
- Bhattacharyya and Hellinger Squared distances performed very similarly
  - ❖ These distances gave the same clusters in some instances, depending on the specified linkage
- Average KL Divergence did not distinguish artist's style well
  - ❖ This measure resulted in very low dissimilarity between all observations, giving little useful information